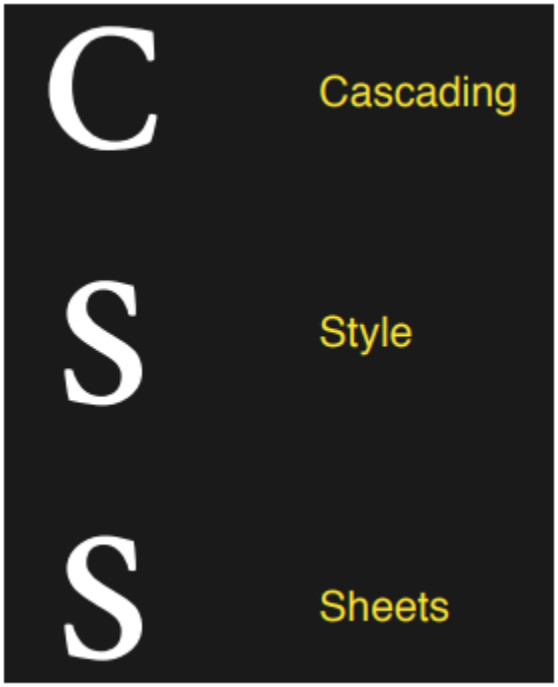
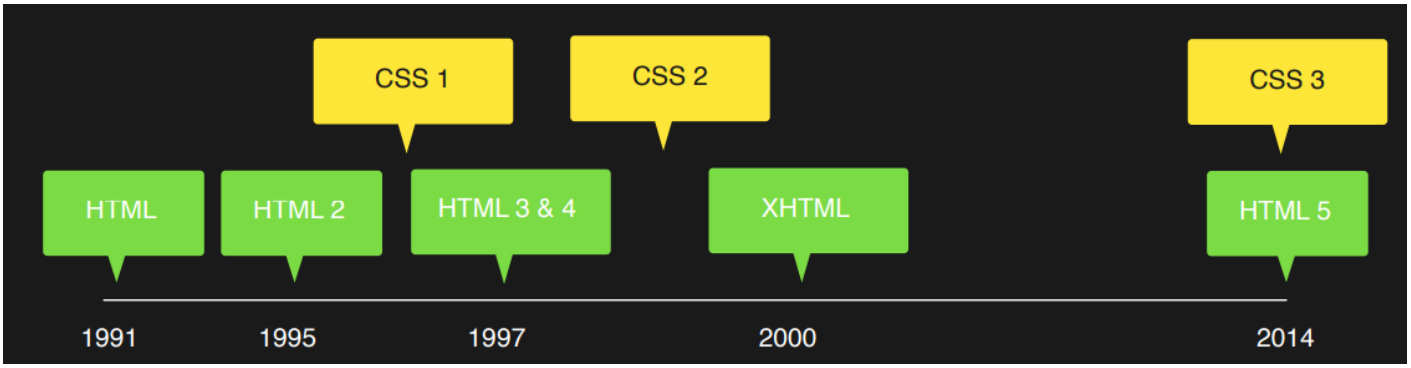
S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development





HIRA KUMAR

❖ Display Property

1. Flex /Flexbox (1d)

- display: flex is a value of the CSS display property that turns an element into a flex container, allowing you to easily align and distribute space among its child elements.
- Parent & child use always for display:flex;

```
<div>
<p> one </p>
</p> two </p>
</p> three </p>
</div>
```

Here is parent & <p> will be child.

(display:flex तभी काम करेगा जब अपने parent और child दोनों लिखे हैं, यदि अपने सिर्फ child लिखा हैं तो flex / flex-direction काम नहीं करेगा)

- "display:flex is used to create a flexible layout system, where elements can be arranged in row or column and aligned easily."
- All items are arranged horizontally (left to right) i.e. row

➢ Why do we use Flexbox?

✓ To center items (Items को center करने के लिए)

✓ To distribute space evenly (Space बराबर distribute करने के लिए)

✓ To create responsive designs (Responsive design बनाने के लिए)

✓ To control layout in both row and column directions (Row और column दोनों control करने के लिए)

- Syntax –

```
{
Display:flex;        /* by default row */
}
```

(element को block-level flex container बनाता है)

2. Inline-flex

- Inline-block is a CSS display value that combines features of both inline and block elements.
- Syntax – {
  Display:inline-flex;
  }

(element को inline flex container बनाता है)

If you want in column (Direction Control) then –

```
{
Display:flex;
flex-direction: row;     /* side by side */
flex-direction: column;  /* upar se niche */
}
```

(यहाँ कोई एक direction ही use करें)

- Example-

```
    <style>
   .a1{
    display:flex;
    flex-direction:row;
   }
   .p1{
    display:inline-flex;
    flex-direction: row;
   }
    </style>
</head>
<body>
    <h3> a tag </h3>
    <div class="a1">
    <a style="background-color: red;"> Home </a>
    <a style="background-color: blue"> result </a>
    <a style="background-color: red;"> Admit card </a>
    <a style="background-color: blue;"> latest job </a>
    <a style="background-color: red;"> Admission </a>
    <a style="background-color: blue;"> Syllabus </a>
    <a style="background-color: red;"> Answer key </a>
</div>
<br><hr>
<h3> p tag </h3>
<span class="p1">
  <p style="background-color: red;"> Home </p>
    <p style="background-color: blue"> result </p>
    <p style="background-color: red;"> Admit card </p>
    <p style="background-color: blue;"> latest job </p>
    <p style="background-color: red;"> Admission </p>
    <p style="background-color: blue;"> Syllabus </p>
    <p style="background-color: red;"> Answer key </p>
</span>
```

Use- row

http://127.0.0.1:5500/2nd pro/ok.html

a tag

Homeresult Admit cardlatest jobAdmissionSyllabusAnswer key

---

p tag

Homeresult Admit cardlatest jobAdmissionSyllabusAnswer key

use-column

```html
<style>
.a1{
 display:flex;
 flex-direction:column;
}
.p1{
 display:inline-flex;
 flex-direction:column;
}
</style>
</head>
<body>
    <h3> a tag </h3>
    <div class="a1">
    <a style="background-color: red;"> Home </a>
    <a style="background-color: blue"> result </a>
    <a style="background-color: red;"> Admit card </a>
    <a style="background-color: blue;"> latest job </a>
    <a style="background-color: red;"> Admission </a>
    <a style="background-color: blue;"> Syllabus </a>
    <a style="background-color: red;"> Answer key </a>
</div>
<br><hr>
<h3> p tag </h3>
<span class="p1">
  <p style="background-color: red;"> Home </p>
    <p style="background-color: blue"> result </p>
    <p style="background-color: red;"> Admit card </p>
    <p style="background-color: blue;"> latest job </p>
    <p style="background-color: red;"> Admission </p>
    <p style="background-color: blue;"> Syllabus </p>
    <p style="background-color: red;"> Answer key </p>
</span>
</body>
```

http://127.0.0.1:5500/2nd pro/ok.html

a tag

Home
result
Admit card
latest job
Admission
Syllabus
Answer key

---

p tag

Home

result

Admit card

latest job

❖ Justify-content Property

It align flex/grid items along the main axis (horizontally by default in flexbox). It controls how extra space is distributed b/w or around items.

Value	Effect
Flex-start (default)	Items packed at the Start (left in LTR)
Flex-end	Items packed at the end (right in LTR)
Center	Items packed in the mid
Space-between	First item at start,last at end, equal space between items
Space-around	Equal space around each items
Space-evenly	Completely equal space b/w and around all items
Stretch	Items stretch to fill container

Value	Example
Flex-start	[A] [B] [C]_____
Flex-end	_____[A] [B] [C]
Center	_____[A] [B] [C]_____

Space-between	[A]_____[B]_____[C]
Space-around	___[A]_____[B]_____[C]_____
Space-evenly	___[A]_____[B]_____[C]_____

Justify-content काम करेगा केवल display:flex या फिर display:grid के साथ Example -

```
    <style>
   .a1{
    display:flex;
    flex-direction:row;
    justify-content:center;
   }
   .p1{
    display:inline-flex;
    flex-direction:column;
   }
    </style>
</head>
<body>
    <h3> a tag </h3>
    <div class="a1">
    <a style="background-color: red;"> Home </a>
    <a style="background-color: blue"> result </a>
    <a style="background-color: red;"> Admit card </a>
    <a style="background-color: blue;"> latest job </a>
    <a style="background-color: red;"> Admission </a>
    <a style="background-color: blue;"> Syllabus </a>
    <a style="background-color: red;"> Answer key </a>
</div>
```

http://127.0.0.1:5500/2nd pro/ok.html 110%

a tag

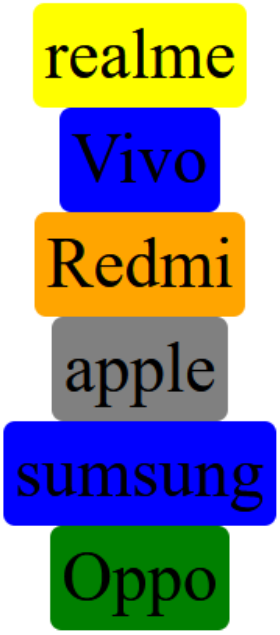
Homeresult Admit cardlatest jobAdmissionSyllabusAnswer key

Next Example -

```
        .container{
            display:flex;
            width:100%;
             justify-content:space-around;
            align-items:center;
            flex-direction:row; /* not mandatory */
        }
    </style>
</head>
<body>
   <div class="container">
    <div class="realme" > Realme </div>
    <div class="vivo"> Vivo </div>
    <div class="redmi"> Redmi </div>
    <div class="apple"> apple </div>
    <div class="sum"> sumsung </div>
    <div class="oppo"> Oppo </div>
   </div>
</body>
```

use-column

```
    <style>
        }
        .container{
            display:flex;
             width:100%;
              justify-content:space-around;
             align-items:center;
             flex-direction:column;

        }
    </style>
</head>
<body>
   <div class="container">
    <div class="realme">realme</div>
    <div class="vivo"> Vivo </div>
    <div class="redmi"> Redmi </div>
    <div class="apple"> apple </div>
    <div class="sum"> sumsung </div>
    <div class="oppo"> Oppo </div>
   </div>
```



❖ Align-item Property

Align-item is a CSS property used in Flexbox and grid layouts to control how items are positioned along the cross-axis.

Value	Effect
Stretch (default)	Items expand to fill the container
Flex-start	Items align at the Top (row) or left (column)
Flex-end	Items align at the bottom (row) or right (column)
Center	Items align at the middle
Baseline	Items align by their text line

Align-item काम करेगा केवल display:flex या फिर display:grid के साथ

3. Grid (2d)

- Grid is used to create rows and columns layout (2D layout system) i.e. as a Matrix.

  1 2 3
  4 5 6

- Syntax – {

  Display:grid;

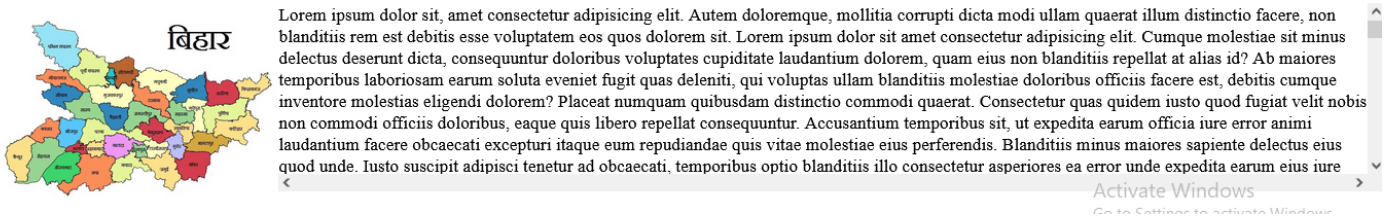
  }

- Example-

```html
<style>
    }
    .container{
        display:grid;
         width:100%;
          justify-content:space-around;
         align-items:center;
          gap:10px;
    }
</style>
</head>
<body>
  <div class="container">
   <div class="realme">realme</div>
   <div class="vivo"> Vivo </div>
   <div class="redmi"> Redmi </div>
   <div class="apple"> apple </div>
   <div class="sum"> sumsung </div>
   <div class="oppo"> Oppo </div>
  </div>
</body>
```

Gap property का use item के बीच space देने के लिए-

```html
<style>
 }
 .p1{
  display:inline-flex;
  height:200px;

 }
 </style>
</head>
<body>

<span class="p1">
  <p style="background-color: red;"><img src="bihar.jpg" width="250px";></p>
   <p style="overflow:scroll"> Lorem ipsum dolor sit, amet consectetur adipisici
   doloremque, mollitia corrupti dicta modi ullam quaerat illum distinctio facer
   rem est debitis esse voluptatem eos quos dolorem sit. Lorem ipsum dolor sit a
   adipisicing elit. Cumque molestiae sit minus delectus deserunt dicta, consequ
   voluptates cupiditate laudantium dolorem, quam eius non blanditiis repellat a
   maiores temporibus laboriosam earum soluta eveniet fugit quas deleniti, qui v
   blanditiis molestiae doloribus officiis facere est, debitis cumque inventore
   dolorem? Placeat numquam quibusdam distinctio commodi quaerat. Consectetur qu
```



HIRA KUMAR

❖ Grid-template Property

- grid-template is a shorthand property used to define the structure of a grid layout, including rows, columns, and grid areas.
- Grid-template → use → need to define rows and columns together.
- Grid-template-columns → use → Only need to specify the number and width of columns.
- Syntax :-

```
{
    Display:grid;
    grid-template: value;
}
```

OR

```
{
display:grid;
grid-template-columns: value;
}
```

➢ Value – for grid-template-columns

✓ Grid-template-columns: 1fr 1fr; (2 equal column create in block {row automatic create a/c to item)

✓ Grid-template- columns: auto auto auto auto (4 column create a/c to content size)

✓ Grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); (use→ responsive cards that auto-fit)

Use Grid template columns

Photo galleries, Pricing tables, card grids, forms.

```html
    <style>
        }
        .container{
            display:grid;
             grid-template-columns: auto auto;
             width:100%;
              justify-content:space-around;
             align-items:center;
              gap:10px;
        }
    </style>
</head>
<body>
   <div class="container">
    <div class="realme">realme</div>
    <div class="vivo"> Vivo </div>
    <div class="redmi"> Redmi </div>
    <div class="apple"> apple </div>
    <div class="sum"> sumsung </div>
    <div class="oppo"> Oppo </div>
   </div>
```

http://127.0.0.1:5500/shop.html 150%

realme

Vivo

Redmi

apple

sumsung

Oppo

Numerical Concept

1vw = 1% of screen width

Screen width	1vw
1000px	10px
1200px	12px
800px	8px

➢ Value – for gride-template

/*2 rows + 3 columns together*/f

✓ Grid-template: 100px auto / 1fr 1fr 1fr;

/* complete page layout with named areas */

✓ Grid-template:

```
"header header" 80px
"sidebar content" 200px
"footer footer" 60px
/ 150px 1fr;
```

Use grid template

Complex full-page layouts, website structure (header/sidebar/main/footer)

Example –

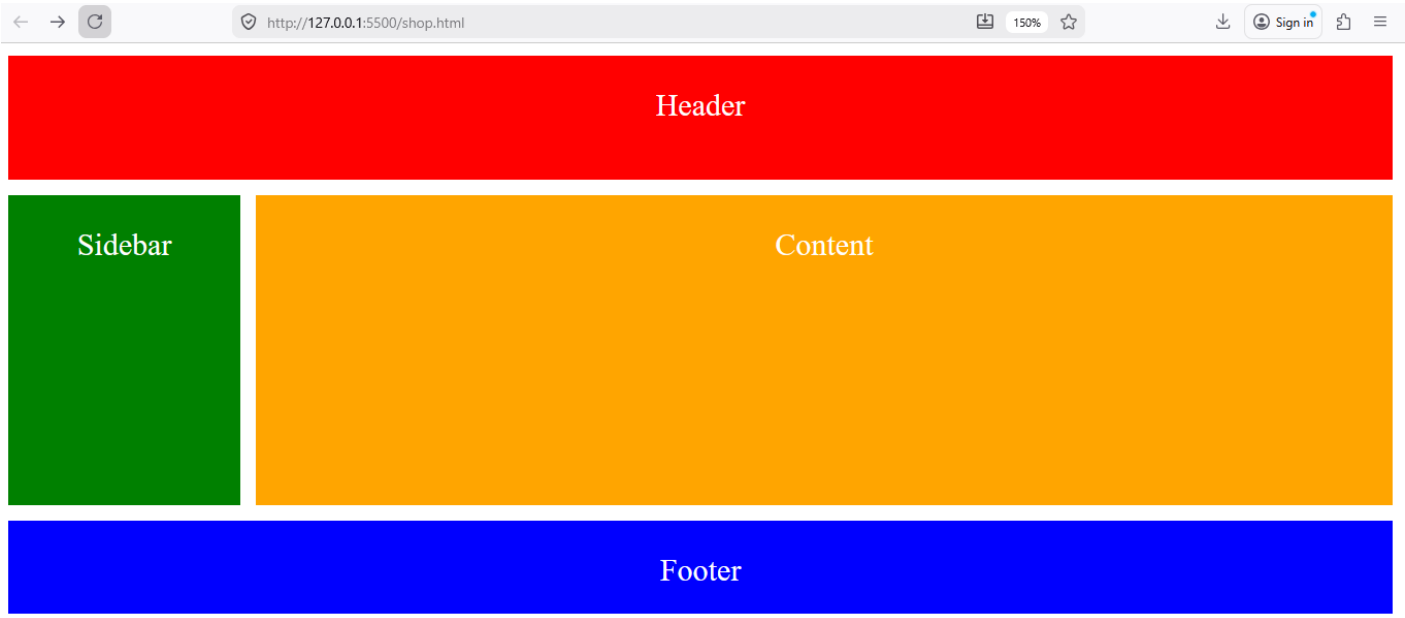
```
<!DOCTYPE html>
<html>
<head>
  <title>Grid Template Example</title>
  <style>
    .container{
      display: grid;

      /* grid-template use */
      grid-template:
        "header header" 80px
        "sidebar content" 200px
        "footer footer" 60px
        / 150px 1fr;

      gap: 10px;
    }
```

Long method

```
grid-template-areas:
  "header header"
  "sidebar content"
  "footer footer";

grid-template-rows: 80px 200px 60px;
grid-template-columns: 150px 1fr;
```

```
        .header{
            background: red;
            grid-area: header;
        }

        .sidebar{
            background: green;
            grid-area: sidebar;
        }

        .content{
            background: orange;
            grid-area: content;
        }

        .footer{
            background: blue;
            grid-area: footer;
        }

        .container div{
            color: white;
            font-size: 20px;
            text-align: center;
            padding: 20px;
        }
    </style>
</head>
<body>

<div class="container">
    <div class="header">Header</div>
```

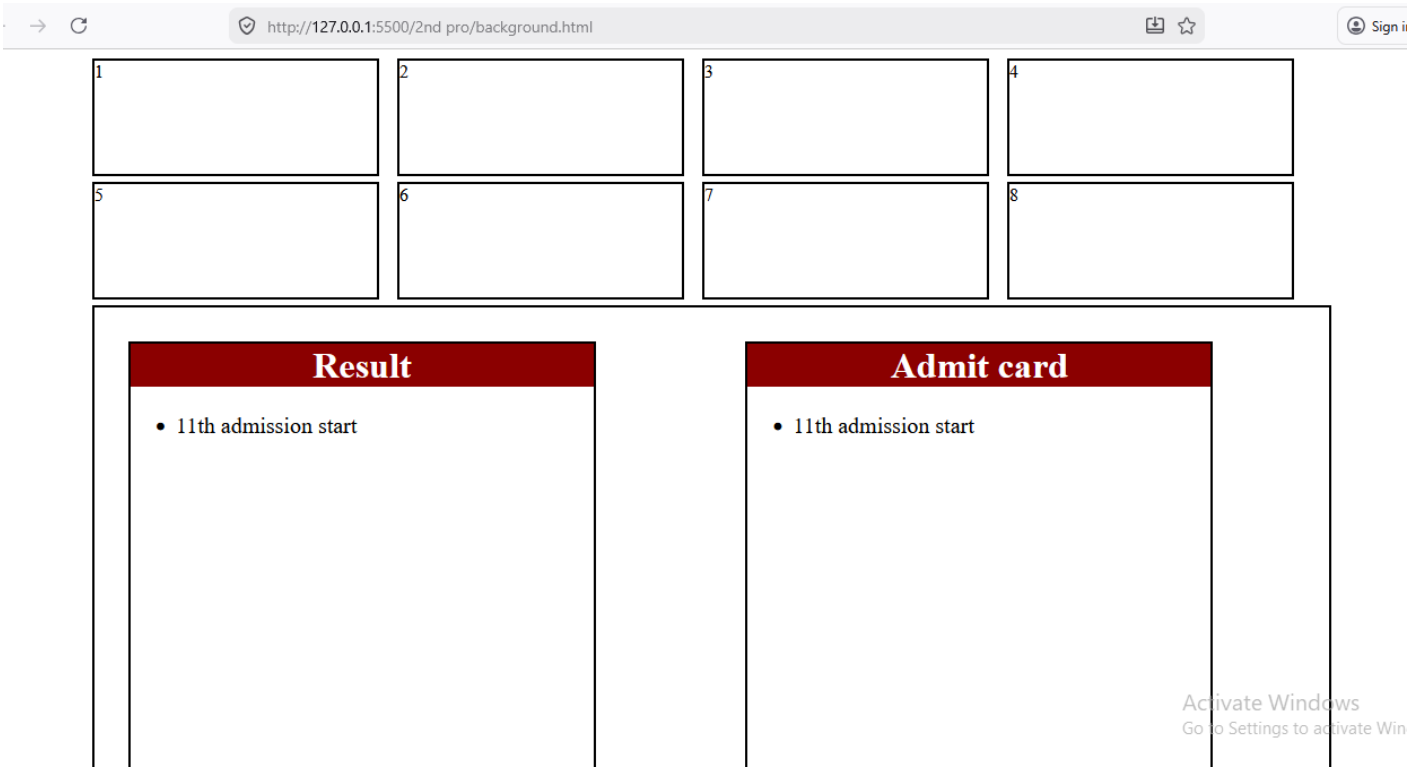
```
        <div class="sidebar">Sidebar</div>
        <div class="content">Content</div>
        <div class="footer">Footer</div>
    </div>

    </body>
    </html>
```



Result bharat-code (flex-grid)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    #id1{
      width:80vw;
      height:150vh;
      margin-left:100px;
      border:2px solid black;
```

```
        display:grid;
        Grid-template-columns:1fr 1fr;

    }
    .class1{
border:2px solid black;
width:30vw;
        height:140vh;
        margin:30px;
    }
    h1{
        text-align:center;
        background-color:darkred;
        color:white;
        margin-top:0px;
    }
    ul li{
        font-size:20px;
    }
    .hira{
        width:250px;
        height:100px;
        border:2px solid black;
        margin-bottom:5px;


    }
    .parent{
        display:grid;
        grid-template-columns: 270px 270px 270px 270px;
        margin-left:100px;

    }
  </style>
</head>
<body>
  <div class="parent">
    <div class="hira">1</div>
```

```
    <div class="hira">2</div>
    <div class="hira">3</div>
    <div class="hira">4</div>
    <div class="hira">5</div>
    <div class="hira">6</div>
    <div class="hira">7</div>
    <div class="hira">8</div>
 </div>
 <div id="id1">
<div class="class1" >
 <h1>Result</h1>
 <ul>
    <li> 11th admission start </li>
 </ul>
</div>
<div class="class1" >
 <h1>Admit card</h1>
 <ul>
    <li> 11th admission start </li>
 </ul>
</div>
<div class="class1">
 <h1>Online</h1>
 <ul>
    <li> 11th admission start </li>
 </ul>
</div>
<div class="class1" >
 <h1>Result</h1>
 <ul>
    <li> 11th admission start </li>
 </ul>
</div>
<div class="class1" >
 <h1>Admit card</h1>
 <ul>
    <li> 11th admission start </li>
 </ul>
```

```
  </div>
  <div class="class1">
   <h1>Online</h1>
   <ul>
      <li> 11th admission start </li>
   </ul>
  </div>
  </div>
</body>
</html>
```